Finite Magma Exercise

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Problem 1. Let M be a finite nonempty set with a binary operation $\diamond$ satisfying

$$x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big) \qquad (x, y \in M). \tag{H}$$

Prove

$$x = ((x \diamond x) \diamond x) \diamond x \qquad (x \in M). \tag{G}$$

Hints

1. For $a \in M$, set $L_a(t) = a \diamond t$. From (H),

$$b = a \diamond \big(b \diamond ((a \diamond b) \diamond a)\big).$$

Hence L_a is bijective. Write $a \backslash b$ for the unique u with $a \diamond u = b$;

$$a \backslash b = b \diamond ((a \diamond b) \diamond a). \tag{D}$$

2. For $y, z \in M$,

$$z = (y \diamond z) \diamond \big((y \diamond (y \diamond z)) \diamond y\big), \tag{T}$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond \big(((y \diamond z) \diamond z) \diamond (y \diamond z)\big). \tag{K}$$

Use (H) with $x = y \diamond z$, then compare with (H) after $y \mapsto y \diamond z$.

3. Fix $x \in M$ and put

$$a = x \diamond x, \qquad q = a \diamond x = ((x \diamond x) \diamond x), \qquad p = x \backslash x.$$

The goal at x is

$$q \diamond x = x. \tag{1}$$

If $u \diamond x = x$, then $u = q$. Also (1) is equivalent to each of

$$q \backslash x = x, \qquad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \qquad (q \diamond x) \diamond q = p. \tag{2}$$

4. Define

$$H(t) = (x \diamond t) \diamond x, \qquad F(t) = x \diamond (t \diamond x).$$

Prove

$$x \backslash t = t \diamond H(t), \qquad F(L_x(t)) = L_x(H(t)). \tag{3}$$

Thus F and H are conjugate. A fixed point of either map is equivalent to (1), and is unique.

5. Let

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k.$$

Let d be the least period of this L_x-orbit, read subscripts modulo d, and set

$$d_k = c_k \diamond x.$$

$$c_{k-1} = c_k \diamond d_{k+1}. \tag{4}$$

$$p = c_{d-1}, \qquad q = d_1 = c_{d-2}, \qquad p \diamond c_1 = q, \qquad d_j = x \Rightarrow j \equiv d-2 \pmod d. \tag{5}$$

6. If $d = 1$, then (1). If $d = 2$, contradiction. If $d = 3$, write

$$c_0 = x, \qquad c_1 = a = q, \qquad c_2 = p.$$

$$x = a \diamond (p \diamond x), \qquad a \diamond p = x, \qquad p \diamond x = p,$$

then

$$(a \diamond a) \diamond a = p \qquad \text{and} \qquad (a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Then $a \diamond a = a$, hence $p = a$, contradiction. Thus any unresolved x has $d \geq 4$.

7. Assume $q \diamond x \neq x$. Then $u \diamond x \neq x$ for all $u \in M$. Since M is finite, there are $y \neq z$ and $e \in M$ with

$$y \diamond x = z \diamond x = e. \tag{6}$$

From (T),

$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x. \tag{7}$$

Moreover $y \diamond e = z \diamond e \Rightarrow y = z$.

8. Prove the collision lemma

$$\boxed{y \diamond x = z \diamond x = e, \quad y \neq z \quad \Longrightarrow \quad e \diamond x = x.} \tag{C}$$

Then (1) follows. Define

$$\Phi(u, v) = \big(u \diamond v, (u \diamond (u \diamond v)) \diamond u\big).$$

$$\Phi(u, v) = (r, s) \Rightarrow r \diamond s = v. \tag{8}$$

$$\Phi^n(u, v) = (\alpha_n, \alpha_{n+2}), \qquad \alpha_n \diamond \alpha_{n+2} = \alpha_{n+1}. \tag{9}$$

$$t = (y \diamond e) \diamond y = (z \diamond e) \diamond z, \qquad e \diamond t = x, \qquad \Phi(y, x) = \Phi(z, x) = (e, t). \tag{10}$$

Use the first repeated pair in the finite forward Φ-orbit of (e, t) and pull it back through (9). It suffices to force $t = x$.

9. Optional: if $d = 4$, write

$$c_0 = x, \qquad c_1 = a, \qquad c_2 = q, \qquad c_3 = p, \qquad r = q \diamond x, \qquad s = p \diamond x.$$

$$x \diamond a = q, \qquad a \diamond x = q, \qquad a \diamond r = x, \qquad x \diamond q = p, \qquad x \diamond p = x, \qquad p \diamond a = q,$$

and

$$r \notin \{x, a, q\}. \tag{11}$$

If $r = p$, show further that

$$q \diamond a = q, \qquad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \qquad s \notin \{x, a, q, p\}. \tag{12}$$

Final goal. For arbitrary x, prove (C), (1), or any identity in (2). Then (G) holds for all $x \in M$.